

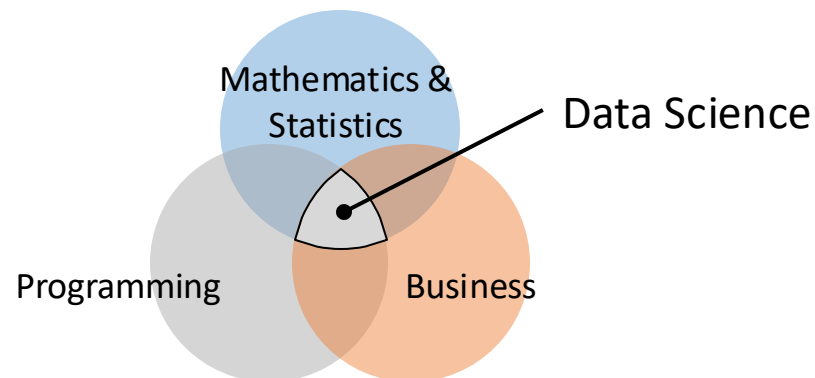
1. What is Data Science?

What is Data Science

A field of study and technology that analyzes large sets of data to provide new insights and information

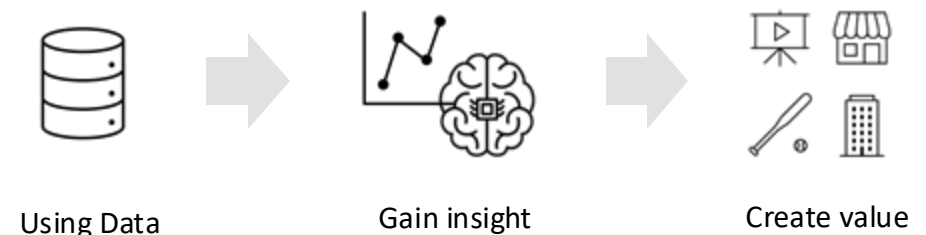
Data Science

- Data Collection and Analysis
Collecting and analyzing various data to gain meaningful insights
- Problem Solving
Making decisions and solving problems through the new insight gained from data
- Interdisciplinary integration
Requires not only programming skills but also knowledge of mathematics, statistics and business.



Example of Data Science

- YouTube Recommendation
Predicting videos users are most likely to watch to increase their time spent on YouTube.
- Predicting Sales of Supermarket
Predicting the sales of each item using historical data, to ensure the right stock at the right time.
- Athlete Recruitment to a Sports Team
Using the data based on the athletes past performances and salaries, the team decides which players to recruit.



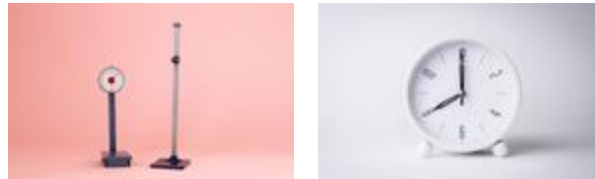
Types of Data

There are two types of data in data science. **Structured data** can be organized in tabular form while **unstructured data** can't be

Types of Data in Data Science

Structured Data

- Organized in rows and columns
 - Quantitative Data
(Numerical and measurable)



- Qualitative Data
(Classification or distinction)



Unstructured Data

- Does not fit into tables

Image



Video



Audio



Text Documents



Most datasets in data science are a combination of quantitative and qualitative data.

Data Collection Methods

Various data exists in the world. However, for self-study, people often use open data or join competitions to get data.

Types of Dataset in the World

Survey Data



Company Data



⋮

Open Data



Competition



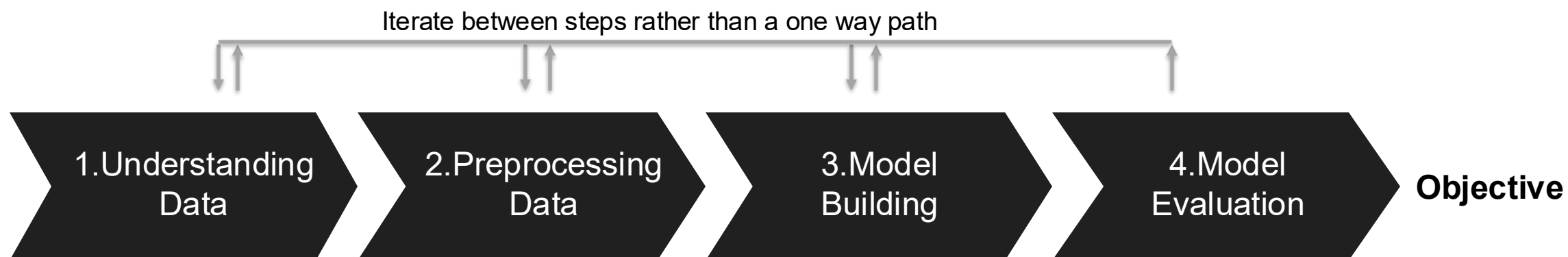
⋮

<Several Data Sources for Learning>

- **Open Data**
Datasets that are freely available online.
 - Government Statistics
 - Public Institution Datasets
- **Competition**
Competitions provide datasets and problems to solve using data analysis and modeling and compete on their results
 - Kaggle
 - SIGNATE

Data Science: Workflow

Data science process consists of four steps (**understanding the data, preprocessing, model building, and model evaluation**)



Overview

- | | | | |
|---|---|--|---|
| <ul style="list-style-type: none"> • Check data distributions • Examine missing values • Hypothesize which data is important for the objective | <ul style="list-style-type: none"> • Organize necessary data • Handling missing/abnormal values | <ul style="list-style-type: none"> • Train a model using data | <ul style="list-style-type: none"> • Evaluate how well the model achieves • (If the objective is not met, return to the previous steps) |
|---|---|--|---|

<Automobile Company Case>

Example

- | | | | |
|---|---|---|---|
| <ul style="list-style-type: none"> • Check for missing data or bias to perform car price prediction • Hypothesize about which factors determine price | <ul style="list-style-type: none"> • Removing irrelevant data not related to hypothesized factors • Handling missing values | <ul style="list-style-type: none"> • Training a car price prediction model using preprocessed data | <ul style="list-style-type: none"> • Compare the predicted price to actual price • (Since there is a significant discrepancy, revisit hypothesis) |
|---|---|---|---|

1. Understanding the Data

Calculate statistical metrics and visualize the data to grasp the overall structure. Hypothesize about whether the data is useful for analysis.

Methods

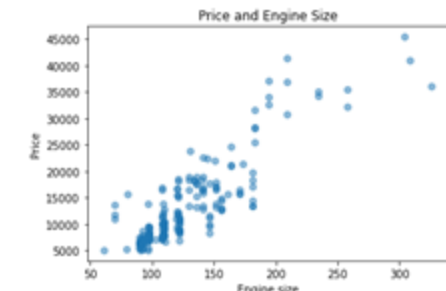
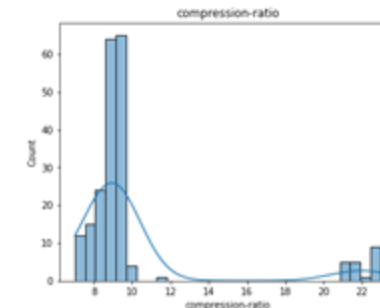
- Compute Basic Statistical Metrics
e.g. Mean, median, variance, etc...
- Visualization
Use graphs and plots to make the data easier to understand.



Through these methods, hypothesize
which data is relevant
→ This process is called understanding
the data

Example Use Case

- There are 21 manufacturers and 6 engines types.
- The compression-ratio data is skewed



The visualizing revealed a trend:

- Trend Observed: Cars with larger engines tend to have higher prices

Hypothesis:

- Engine size may be related to price.

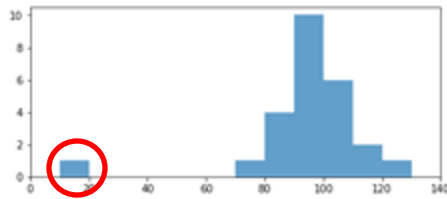
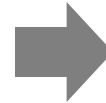
Data needs to be preprocessed into a format that is easier to analyze. This mainly involves cleaning and transforming the data.

Cases Where Preprocessing is Needed

- Outliers or Missing Values

Outliers : Values that are extremely different from the rest of the data

Missing Values : Values that are missing or not recorded.

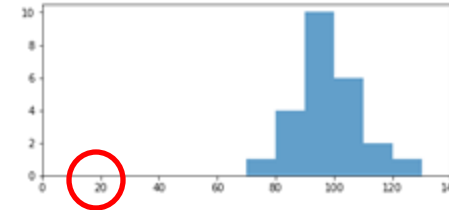


Name	Height
Jack	
John	160

Preprocessing Method

- **Data Cleaning**

Replace missing or outlier data values with alternatives such as the average or remove them



Name	Height
John	160

Data needs to be preprocessed into a format that is easier to analyze. This mainly involves cleaning and transforming the data.

Cases Where Preprocessing is Needed

- Cases where scales of feature differ
Scale: the range or unit of a variable's values



Preprocessing Method

- Standardization**
Adjust so that the mean is 0 and the variance is 1

*Mean, variance, standard deviation are explained later

- Dummy Encoding**
Convert categorical data into 0 or 1

- Cases with categorical data

Office	Department
Head Office	Sales
Branch	R&D
Head Office	HR



- Office
Head $\Rightarrow$ 0、Branch $\Rightarrow$ 1
- Department
Sales $\Rightarrow$ 1 0、R&D $\Rightarrow$ 0 1
HR $\Rightarrow$ 0 0

Office	Sales	R&D
0	1	0
1	0	1
0	0	0

A **model** is a set of computational rules devised to meet a specific objective. Constructing such a model typically involves three steps: **Data Splitting, Model Selection, and Model Training.**

Overview of Model

A set of computational rules built to achieve a specific objective.

E.g.) To understand the relationship between the x and y-axes, a linear model like $y = ax + b$ can be used:

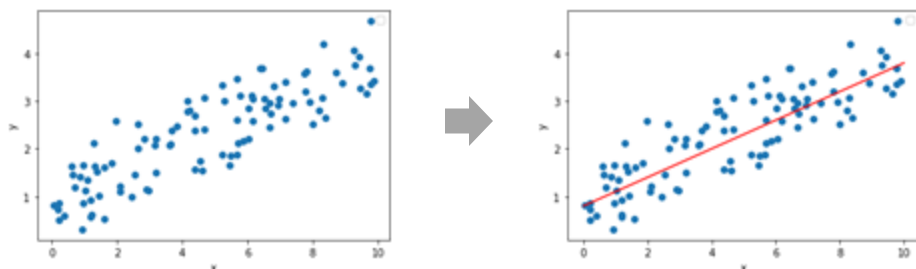
$$y = ax + b$$

y : Value to predict (Output)

x : Value used to predict (Input)

a, b : Parameters (Model learns them)

<Visual Interpretation>



Building a Model

Process

①
**Data
Splitting**

②
**Model
Selection**

③
**Model
Training**

Details

- Split the dataset into training and test sets before evaluating the model. This allows you to test the model from different perspectives.
- Choose a model that fits the data and aligns with the objective. You can also try different models based on your assumptions.
- Train the selected model using the training data to learn patterns and make predictions.

Data splitting is used to evaluate how well a model generalizes to unseen data. **Hold-out method & k-fold cross-validation** are often used

Purpose of Data Splitting

Objective of the model = Predict unseen data

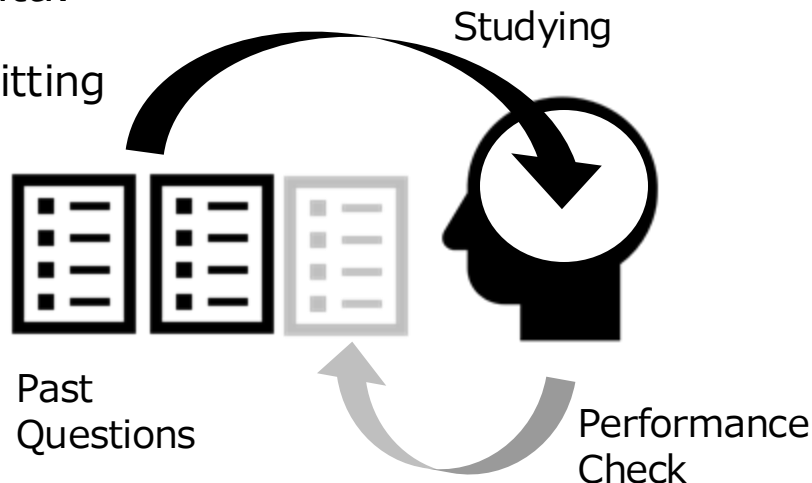
- **Data Splitting**

- Instead of using all the data to train the model, use a portion set aside as test data. This allows us to evaluate how well the model performs on new, unseen data.

E.g. Data Splitting

Objective

Passing the Test



Methods of Data Splitting

- **Hold-out Method**

Splitting the dataset into one pattern



- **k-Fold Cross-Validation**

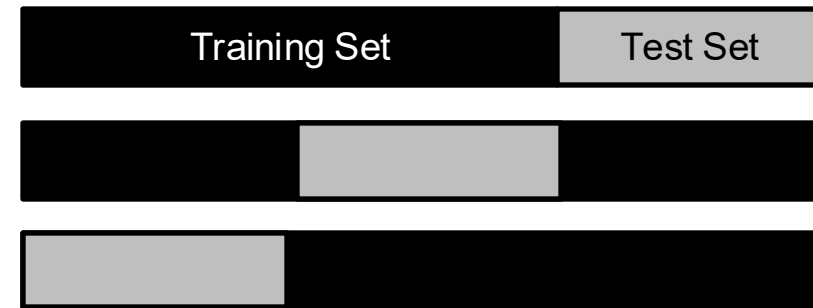
Splitting data into k different patterns

Pros : Reduce bias from single test set

Cons : Higher computational cost

(multiplied by k)

e.g.) If $k=3$



Model selection means choosing a model that fits the data and aligns with the objective.

Overview of Model Selection

Model Selection

=Choosing the appropriate model based on the objective

- **Examples of objectives**

- **Regression**

- Predicting Numerical Values
(e.g. Predicting house prices)

- **Classification**

- Categorizing data into different classes
(e.g. Identifying if email is a spam)

- **Clustering**

- Grouping data into several clusters
(e.g. Customer segmentation)

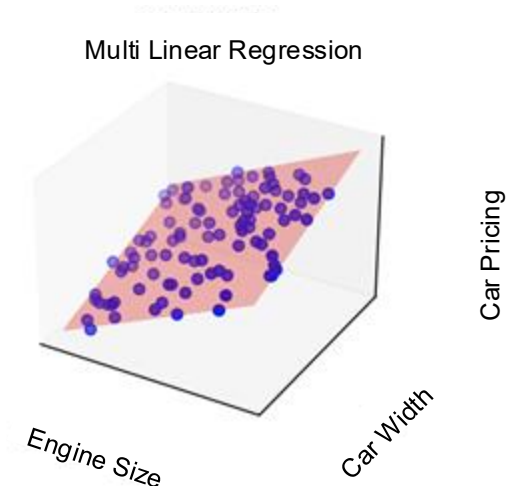
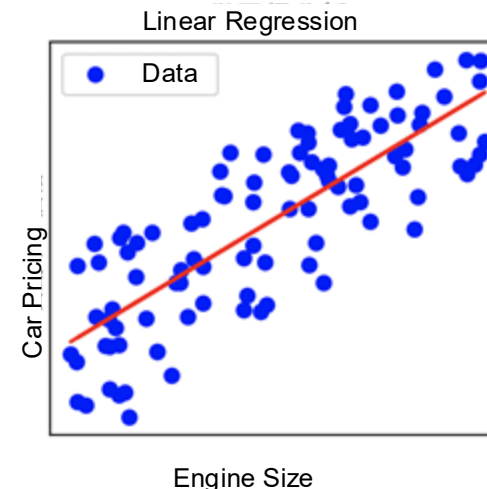
Examples of Model Selection

Objective: Predict car prices

→ Since the objective is to predict a numerical value, a regression model is appropriate

Linear Regression : One variable to predict

Multiple Linear Regression : Multiple variables to predict



Model training means training the selected model using the training data to learn patterns and make predictions.

Overview of Model Training

Model Training

= Building a machine learning model using training data

Training Data

Test Data

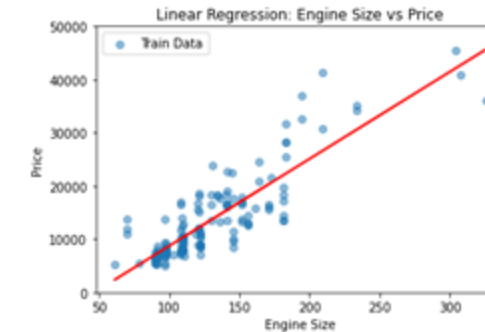
Example of Model Training

Linear Model: $y = ax + b$

y : Value to predict (Output) → Car's Price

x : Value used to predict (Input) → Engine Size

a, b : Parameter (Model learns them)



Using the training data, the model learns and estimates the parameters a and b that best fit the data in the form of $y = ax + b$

➡ $a : 163.7$ $b : -7664.5$

If Engine Size is 100,

$y = 163.7 \times 100 - 7664.5 = 8705.5$ is the Car's Price

Model evaluation is the process of assessing how well a trained model achieves its objective and how reliable it is.

Overview of Model Evaluation

Model Evaluation

= Measuring how accurately a trained model can predict unseen data

By using appropriate evaluation metrics, we can:

- Evaluate a model's performance
- Compare different models

Note:

The evaluation metrics vary depending on the model type.

Examples of Evaluation Metrics

- **Regression Model**

- Mean Squared Error (MSE)

The avg of squared differences between predicted and actual values

- Always ≥ 0
- Smaller MSE $\rightarrow$ More accurate

- **Classification Models**

- Accuracy

The proportion of data points correctly classified by the model

- Value ranges from 0 to 1
- If closer to 1, more accurate

Mean Squared Error (MSE) is a common evaluation metric for regression models, while accuracy is commonly used for classification

Example of Evaluation Metric ①

- Predicting Car Prices
 - Mean Squared Error

	Actual Price	Predicted Price	Error	Squared Error
A	47000	46400	600	360000
B	20000	21500	1500	2250000
C	32000	32900	900	810000

$$\frac{360000 + 2250000 + 810000}{3} = 1140000$$

MSE is 1140000

※Since MSE is affected by the scale of the target variable (e.g. car prices), it is best used for comparison between models on the same dataset.

Example of Evaluation Metric②

- Classifying Mushrooms as Poisonous or Safe
 - Accuracy

	Actual : Poisonous	Actual : Safe
Predict : Poisonous	35	10
Predict : Safe	5	50

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

$$= \frac{35 + 50}{100} = \frac{85}{100} \quad \text{Accuracy is 0.85}$$

Return to earlier steps and explore ways to improve or try alternative approaches.

Quiz - Check Your Understanding



Choose the correct words for each step from the options below.

Review of the Data Science Workflow



Options :

Data Preprocessing, Model Evaluation, Model Building, Data Understanding

Quiz - Check Your Understanding



Choose the correct words for each step from the options below.

Review of the Data Science Workflow

